Predicting Airborne Disease Spread Using Urban Mobility and Environmental Factors

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Background

Airborne dieases can be predicted by models trained on historical cases and urban mobility indices.
We investigate whether including additional environmental factors in our model can improve the prediction accurancy for airborne disease (e.g. flue, Covid19) spread.

Environmental factors taken into consideration :

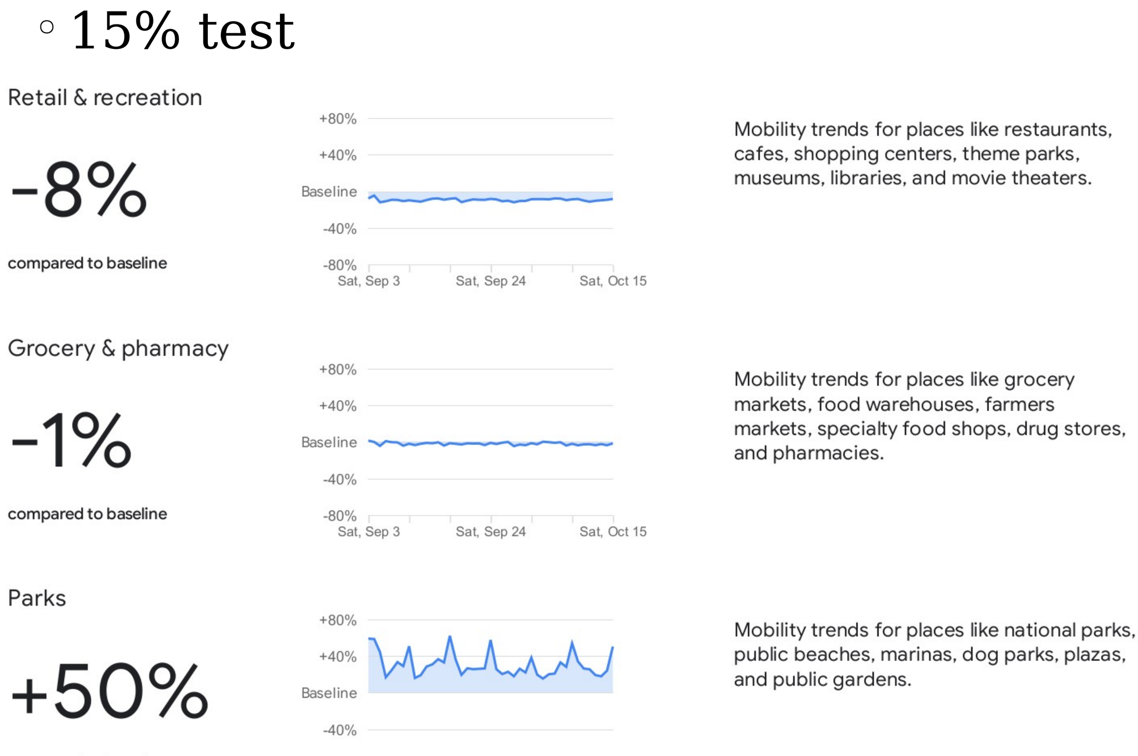
- Temperature [1]
- Humidity [1]
- pollution indices (PM2.5, Ozone) [2]

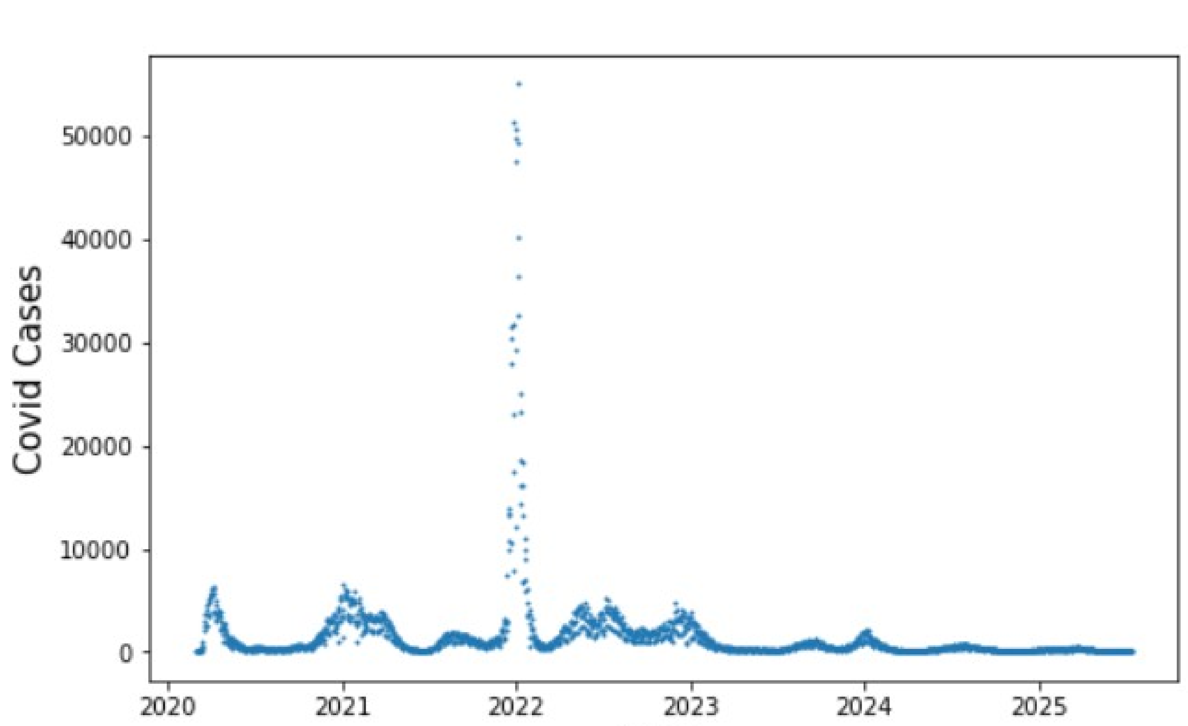
Dataset

To investigate which factors lead to a higher prediction accuracy, we compiled a data set of public available mobility reports [3], covid19/ flu data [4] , and air quality data [1,2] for New York City.

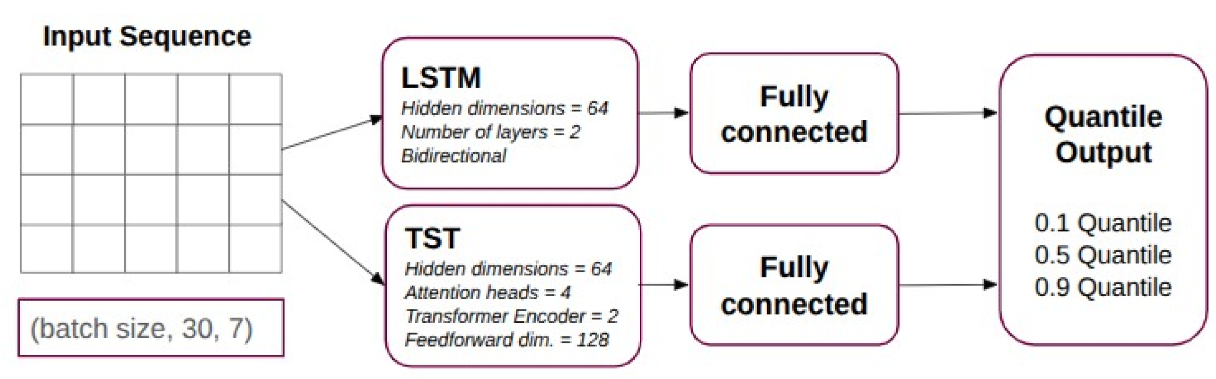
Our dataset was split:

- 70% training
- 15% validation
- 15% test





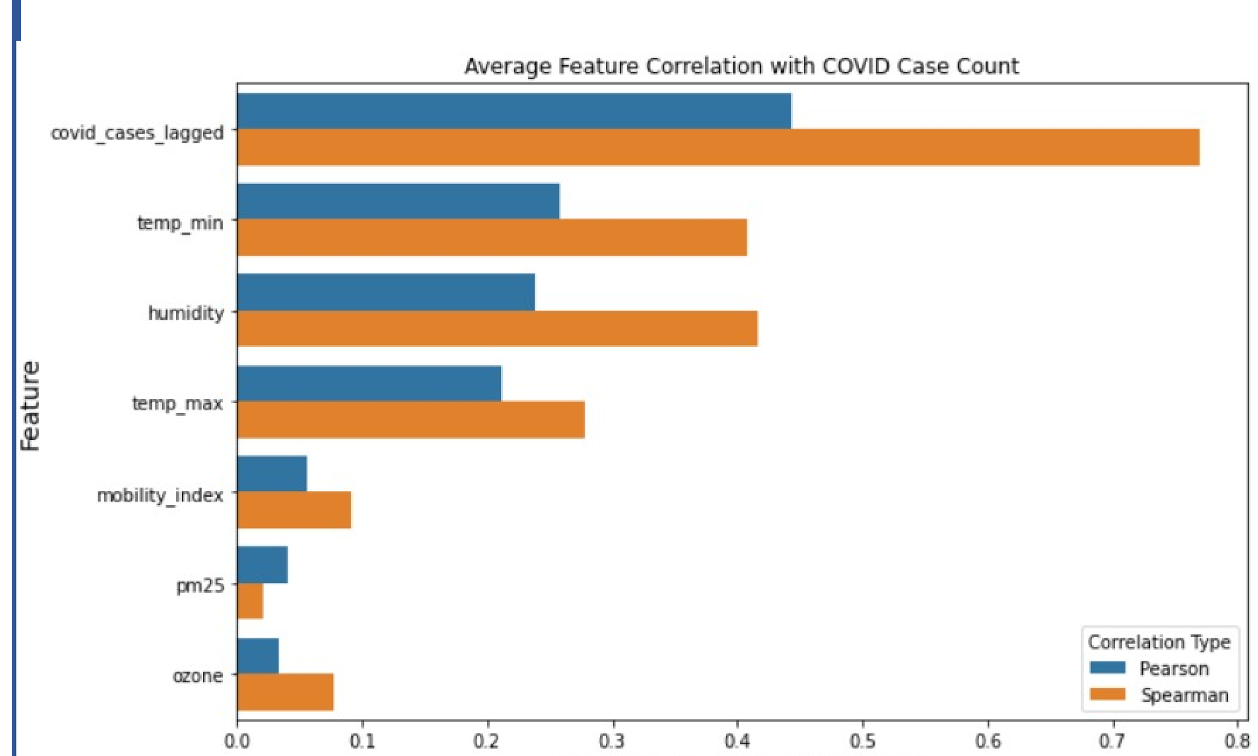
Model



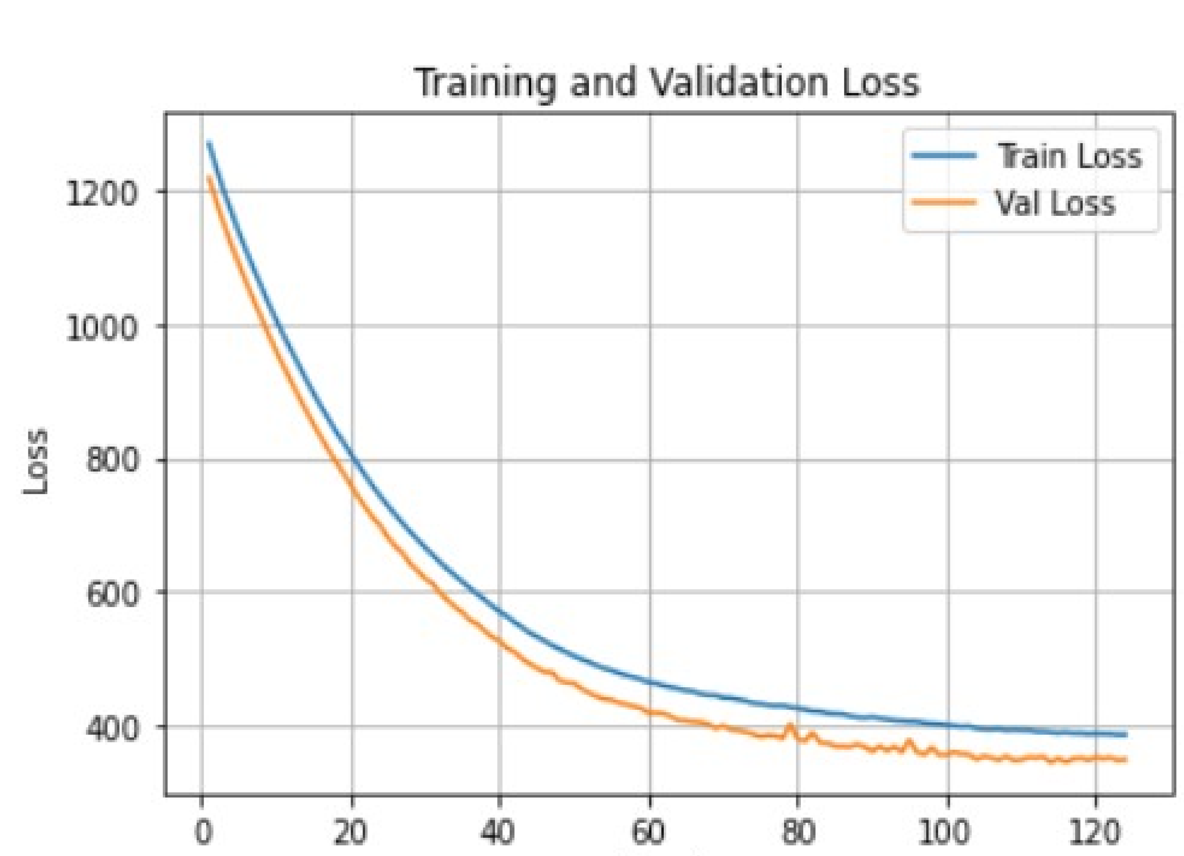
Different deep learning architectures were considered and the two different models tested with one time a TimeSeries Transfoermer (TST) and a Long Short Term Memory (LSTM):

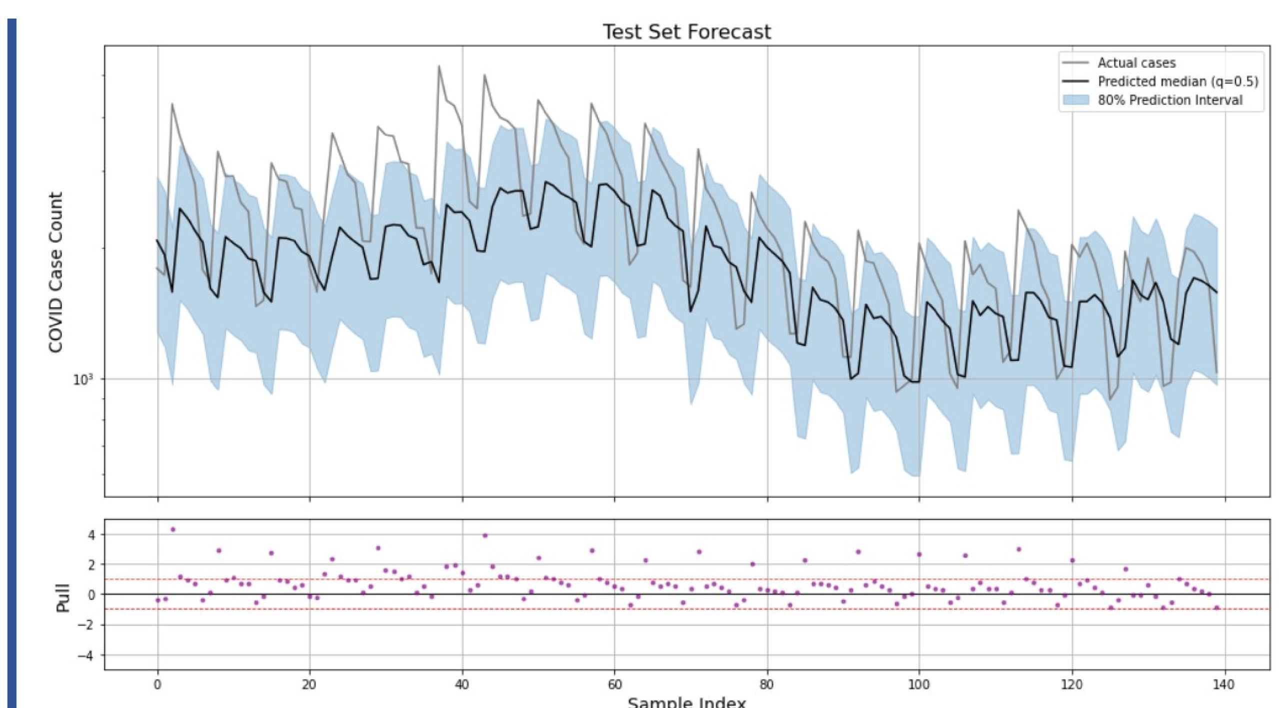
- Use an LSTM or Transformer model to process 30 days of past data
- Predict the 10th, 50th, and 90th percentiles of the next-day case count
- Use pinball loss to train on quantiles (not just the mean)

Features: mobility_index, PM2.5_AQI, Ozone_AQI, temp_max, temp_min, humidity, covid_cases



Results





How confident can we be in our predictions?

Our 80% prediction band (between 10th and 90th quantile) covered the actual cases 75%

Can mobility + air quality explain spatial clusters of disease spread better than historical cases alone?

Yes

Next Steps

- Extend the model to other cities
- Multi-day forecast
- Train for additional airborne diseases

References

1 - Open-Meteo. (2024). *Weather and climate data API*. Retrieved from https://archive-api.open-meteo.com/v1/archive

2 - U.S. Environmental Protection Agency. (2022). *Air Quality System (AQS) Data Mart*. Retrieved from https://aqs.epa.gov/aqsweb/airdata/download_files.html

3 - Google LLC. (2022). *COVID-19 Community Mobility Reports*. Retrieved from https://www.google.com/covid19/mobility/

4 - NYC Department of Health and Mental Hygiene. (2024). *COVID-19 daily case data*. Retrieved from https://github.com/nychealth/coronavirus-data